WAVE, WATER, AND GREEN

A Coastal Data Center Model for Ethical AI Infrastructure

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Abstract

The current model of data center infrastructure externalizes its costs onto communities least equipped to absorb them. Freshwater is drawn from drought-stressed regions. Power grids serving working-class neighborhoods are strained. Fossil fuel combustion pollutes the air of communities that were never consulted. This is not an inevitable feature of AI infrastructure. It is a design choice, and design choices can be unmade. This paper proposes a three-component integrated system that inverts that relationship: the Waveline Magnet wave energy converter, coastal data center architecture, and a perimeter plant biome for carbon filtration. AI infrastructure can be transformed from a net extractor of community resources into a net contributor. The technology exists or is in final pre-commercial development.

AI Disclosure

AI language models assisted with drafting and editing. All research, analysis, and conclusions are the work of the human author.

1. The Problem: What Data Centers Currently Cost Communities

In Memphis, Tennessee, Elon Musk's Colossus supercomputer facility — built to train Grok AI models — was powered by 35 methane gas turbines. The surrounding community, a working-class Black and brown neighborhood with an existing history of environmental burden, was not informed before the facility arrived. Residents discovered it when they smelled what they thought was a gas leak in their living rooms. The facility was pumping thousands of tons of toxins into air already carrying elevated rates of lung cancer and respiratory illness.

This is not an anomaly. It's a pattern. When these facilities enter communities, three things reliably happen: power utility costs increase and grid reliability decreases for existing residents; freshwater resources already constrained in drought-affected regions face new industrial competition; air quality degrades through fossil fuel combustion used for primary or backup power. Communities are selected precisely because they offer cheap land, accessible infrastructure, and limited political resistance.

The conditions described here are not an unavoidable consequence of artificial intelligence. They're the result of specific architectural and economic decisions. If those decisions change, the outcomes change. The question is not whether AI will consume resources. It's whether its supporting systems are designed to extract or to reciprocate.

2. Component One: The Waveline Magnet

2.1 What It Is

The Waveline Magnet is presented here as one implementation of a broader class of offshore renewable infrastructure systems. The architectural model in this paper doesn't depend on a single technology succeeding, but on the integration of continuous renewable energy generation, desalination, and thermal exchange into a unified platform.

Sea Wave Energy Limited (SWEL), a research and development company based in Cyprus and the United Kingdom, has been developing the Waveline Magnet since 2006. It's a wave energy converter that harnesses ocean wave motion to produce electricity and freshwater simultaneously. Nearly two decades of R&D;, including live sea deployments and institutional tank testing at Centrale Nantes and the University of Cyprus, underpin its current design.

The device consists of several floating platforms linked in a spine-like configuration, designed to move with waves rather than resist them. SWEL calls this Neutral Displacement Theory. It eliminates the survivability problems that have plagued other wave energy systems. The device can't sink. It requires minimal maintenance. It can be manufactured from recycled and reinforced plastics, dramatically reducing production cost and eliminating contamination risk, since seawater itself is the working fluid.

2.2 Institutional Validation

SWEL's work has been validated across multiple institutional contexts. The technology has been presented at the International Conference on Ocean Energy, the European Wave and Tidal Energy Conference, and tested at Centrale Nantes and the University of Cyprus. The design has received funding or endorsement from the EU's Horizon 2020 programme, the European Regional Development Fund, the UK's Innovate UK, and similar bodies. SWEL's co-CEO was interviewed in Coastal Wave Resource (CWR) on the system's modular design and scalability.

2.3 Output at Scale

At its largest documented configuration, approximately 600 metres in length and 24 metres in width, a Scotland case study estimated output of roughly 140 gigawatt hours per year. An array of ten full-sized units at 200 metres each has been projected to deliver over 15 MW.

For data center application, the relevant question is not single-unit output but array output scaled to the facility's requirements. The system is designed to be deployed as an offshore array, not as a single point source. The freshwater and power math remains viable at that scale.

2.4 The Freshwater Math at Array Scale

A large data center can consume approximately 3 to 5 million gallons of water per day for cooling, equivalent to roughly 11,000 to 19,000 metric tons per day. At an initial commercial unit output of 10 to 50 metric tons per hour, covering a large facility's full cooling requirement would require an array of approximately 10 to 80 units depending on wave conditions and unit configuration. Given the Waveline Magnet's modular design and low per-unit manufacturing cost, this is an engineering and logistics challenge rather than a fundamental constraint.

The array surplus, freshwater beyond cooling requirements, remains a community benefit regardless of array size. The inversion of the community relationship holds at any viable deployment scale.

3. Component Two: The Coastal Data Center

3.1 Siting Logic

Coastal siting becomes a strategic asset in this model. The ocean provides three things simultaneously: the energy input for the Waveline Magnet array, the source water for desalination and cooling, and the disposal pathway for thermal output that would otherwise stress inland air and water systems.

Coastal siting doesn't mean vulnerable community siting. Offshore Waveline Magnet arrays feed power and water to a coastal facility positioned away from residential zones, with community benefit delivered through infrastructure — freshwater supply, grid contribution — rather than proximity.

3.2 Energy and Cooling Architecture

Primary power comes from the Waveline Magnet array offshore. The facility's backup generation becomes a genuine emergency-only system rather than a routine operational component, dramatically reducing direct carbon output at the facility level. Excess power beyond the facility's operational needs can be fed into the local grid.

Cooling water is supplied by the Waveline Magnet's desalination output. Closed-loop cooling systems return water to the desalination cycle rather than expelling it as heated discharge. Thermal output that can't be captured in the cooling loop is directed into the adjacent greenhouse infrastructure, where elevated temperatures support growth rather than causing harm.

Thermal reuse pathways are prioritized where feasible, with dissipation acting as a fallback. Closed-loop cooling and heat reuse significantly reduce environmental impact but don't eliminate thermal output entirely. Excess heat must still be managed through secondary dissipation pathways such as ocean exchange, thermal storage, or industrial reuse. The goal is not perfect closure, but controlled and minimized external impact.

3.3 Community Impact Comparison

Impact Category	Current Model	Integrated Model
Energy source	Grid (fossil fuel mix) or on-site gas turbines	Offshore wave array (renewable, continuous)
Water use	Draws from local freshwater supply	Desalinated seawater; surplus returned to community
Cooling discharge	Heated water expelled to local environment	Closed-loop reuse; thermal directed to biome
Air quality	Fossil combustion degrades local air	Biome filtration; backup gen as emergency only
Community relation	Extractive; no direct benefit	Reciprocal; water, power, green space returned

Table 1. Community impact comparison: current data center model vs. integrated coastal model.

4. Component Three: The Perimeter Plant Biome

4.1 The Science of Plant-Based Air Filtration

The plant biome is not intended as a primary solution to global emissions. Its role is local: improving air quality, stabilizing microclimates, and contributing to human well-being in proximity to the infrastructure. Primary environmental impact reduction comes from the energy and cooling systems. The biome is a complementary layer.

Plants convert carbon dioxide into oxygen through photosynthesis. Beyond photosynthesis, plant leaves and their associated root-zone microorganisms absorb and metabolize a range of airborne pollutants including carbon monoxide, nitrogen oxides, sulfur dioxide, volatile organic compounds, benzene, formaldehyde, and particulate matter. Species diversity is a key design consideration since different plants target different pollutants.

4.2 Living Wall Architecture

Living walls, vertical gardens integrated into building exteriors, are an established architectural technology with documented air quality benefits. They transform bare structural walls into active biological filtration systems without requiring additional land footprint. In high-density environments, they provide the only viable pathway to increase meaningful plant surface area where horizontal land is unavailable or prohibitively expensive.

4.3 Design Concept for Data Center Application

The perimeter plant biome serves multiple simultaneous functions:

Carbon and pollutant filtration: Species selected for documented absorption capacity of carbon monoxide, NOx, SOx, particulate matter, and VOCs, positioned at the facility perimeter to capture residual exhaust and improve ambient air quality.

Thermal buffer and greenhouse integration: Waste heat from the data center becomes the controlled growing environment for the biome, supporting year-round productive plant growth without additional heating infrastructure.

Freshwater integration: Surplus desalinated water from the Waveline Magnet array provides irrigation, closing the water loop within the integrated system.

Community green space: The biome creates a living boundary between the facility and surrounding area — accessible, productive, community-facing.

Agricultural potential: Greenhouse conditions with reliable water supply create conditions for food production integrated into facility design.

4.4 Caveats on Plant Filtration at Scale

Laboratory results, including NASA's foundational 1989 study, were conducted in sealed chambers under controlled conditions. Real-world outdoor environments involve continuous air exchange that reduces the relative concentration effect of any fixed plant surface area.

This is why the biome is not designed as a sole carbon mitigation strategy. The biome addresses residual exhaust, particulate matter, and NOx from backup generator use, improves ambient air quality at the facility perimeter, and provides the community benefit layer that changes the relationship between the facility and its neighbors. The biome's full impact at data center perimeter scale requires dedicated research to quantify.

5. The Integrated System

5.1 System Summary

The three components form a closed-loop system where each element addresses the waste outputs of the others:

The Waveline Magnet array converts ocean wave energy into electricity and freshwater, eliminating fossil fuel grid dependency and community freshwater competition. The data center consumes both outputs with minimal community impact. Surplus freshwater and power are returned to the surrounding region. The plant biome uses the facility's thermal waste as a growing environment, filters residual carbon and particulate output, and creates community-accessible green space with food production potential.

The defining property of this system is that each output becomes an input to another process. Energy generation powers computation. Waste heat and byproducts are redirected into ecological or industrial use. This doesn't eliminate resource consumption, but it prevents the accumulation of unaccounted externalities — the condition that defines extractive infrastructure.

5.2 Output Specifications

Component	Output	Community Benefit
Waveline Magnet array (10 units, 200m each)	15+ MW electricity; 10-50 MT/hr freshwater per unit	Surplus power to grid; surplus water to community
Data center cooling	Closed-loop water reuse; thermal directed to biome	Reduced discharge impact; greenhouse heating
Plant biome	Air filtration; greenhouse food production	Accessible green space; local food supply

Table 2. Integrated system output specifications and community benefits.

6. What This Is Not

This paper doesn't argue that AI development is inherently beneficial or that building more data centers is the right direction for civilization. Those are separate questions with serious arguments on multiple sides.

This paper argues something narrower: if data centers are going to be built — and they are being built, at accelerating pace, regardless of this or any other argument — then the choice of how to build them is not neutral. The current model is an active choice to externalize cost onto vulnerable communities. This paper proposes a structural alternative that doesn't require that choice.

This paper also doesn't overstate the Waveline Magnet's current development status. SWEL is in final pre-commercial development. Output specifications scale with array size and wave conditions. The freshwater output figures cited reflect currently documented ranges, not maximum theoretical projections. The integrated system is a framework for what responsible infrastructure could look like, not a deployment-ready engineering specification.

This model depends on several conditions: viable offshore energy generation at array scale, effective integration of desalination and cooling systems, and regulatory environments that permit coastal deployment. Where these conditions are not met, the model may not be applicable in its current form.

7. Next Steps

1. **SWEL partnership and specification verification.** Direct engagement with Sea Wave Energy Limited to obtain primary source documentation on freshwater output ranges at array scale and explore data center integration specifications.

2. **Array sizing study.** Detailed analysis of Waveline Magnet array configuration required to meet specific data center power and water requirements at a defined coastal location.

3. **Biome species selection research.** Identification of optimal plant species for outdoor perimeter filtration at data center scale, including real-world rather than sealed-chamber performance data.

4. **Community benefit modeling.** Quantification of freshwater surplus, grid contribution, and green space access for a defined coastal community adjacent to a pilot facility.

5. **Regulatory and consent framework.** Development of a community engagement model that treats surrounding populations as stakeholders in facility design rather than subjects of its impact.

6. **Pilot location analysis.** Identification of a candidate coastal site meeting wave energy criteria, proximate to data center infrastructure needs, with a community that would benefit from freshwater and power surplus.

7. **Staged pilot deployment.** A plausible deployment pathway beginning with a mid-scale pilot facility paired with a limited offshore array, designed to validate integration before scaling to full data center demand.

7.1 Economic and Deployment Considerations

Systems of this kind are not purely environmental proposals. They're also economic structures. Rising regulatory pressure, long-term energy cost stability, infrastructure resilience, and public opposition to extractive development all create incentives for alternative designs. Initial capital costs may be higher, but integrated systems can reduce long-term operational volatility and external risk exposure. Adoption is not solely a matter of ethics. It's strategic positioning within an evolving technological and regulatory landscape.

The Memphis community smelled gas in their living rooms and discovered a supercomputer had moved in next door. Nobody asked them. Nobody offered them anything. The facility took their clean air and gave them nothing back.

That is not the only way to build.

The ocean produces energy constantly. It produces freshwater at scale when the right technology is applied to it. The waste heat a data center produces can grow things instead of baking communities. The perimeter of a facility can be a living biological system instead of a fence.

These are not utopian ideas. They are engineering decisions. The technology exists or is nearing readiness, and the system scales at the array level. What has been missing is a framework that treats communities as something other than a resource to be extracted from.

This paper proposes that framework. The rest is a matter of will.

The trajectory of artificial intelligence will be shaped not only by its algorithms, but by the systems that sustain it. Those systems are still within human control.

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